# 3.3 Linear Functions

Algebra I 3.3 · Student sheet

Name _____ Period ____ Date _____

**I can…**

- tell from a graph or a table whether a function is linear.
- tell from an equation whether a function is linear.
- decide whether a domain is discrete or continuous.

**Vocabulary** · fill in the missing word

**Linear function** — a function whose graph is a _____ line; it has a constant rate of change

**Nonlinear function** — a function without a _____ rate of change

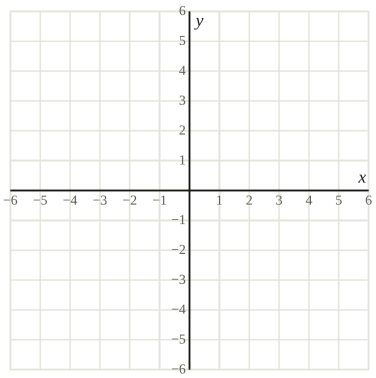
**Discrete domain** — _____ input values (counts, such as people or tickets)

**Continuous domain** — all values in an _____ (measurements, such as time or weight)

## Part 1 · Linear or not?

| x | 0 | 1 | 2 | 3 | 4 | Changes in y | Linear? |
|---|---|---|---|---|---|---|---|
| Table A: y | 2 | 5 | 8 | 11 | 14 | | |
| Table B: y | 1 | 2 | 4 | 8 | 16 | | |
| Table C: y | 0 | 1 | 4 | 9 | 16 | | |
| Table D: y | 10 | 8 | 6 | 4 | 2 | | |

Graph two of the tables:




## Part 2 · Notes: fill in the blanks

A function is linear when its rate of change is _____.

Discrete domain: you can only _____. Continuous domain: you can _____.

$y = x^2$ and $y = 5/x$ are _____.

## Part 3 · Examples (work them with the class)

### Example 1

Is each table linear?

a. x: 0, 2, 4, 6; y: 1, 5, 9, 13

b. x: 1, 2, 3, 4; y: 1, 4, 9, 16

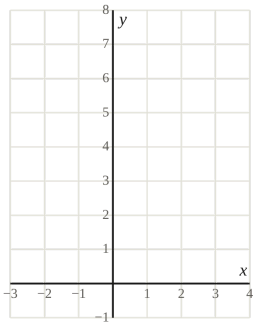
### Example 2

Is each equation linear?

a. $y = 3x - 7$ b. $y = x^2 + 1$ c. $y = \frac{5}{x}$ d. $2x + 3y = 6$

### Example 3

The graph shows two functions. Which one is linear?




### Example 4

Is the domain discrete or continuous?

a. the number of tickets sold and the money collected

b. the gallons of gas pumped and the cost

## Part 4 · Partner practice

### Partner 1

Is each equation linear? a. $y = 4 - 2x$ b. $y = x^3$

### Partner 2

Is the domain discrete or continuous? a. a puppy's weight over time b. the number of people in a car